TRAIN SIM WORLD · VEHICLE WORKFLOW GUIDE

Part 3

Animation: clips scrubbed by values

Nothing on a TSW locomotive plays over time. Wheels are posed from the simulation's axle angle, doors and levers from a control's value, buffers from physics. That is exactly how TSC's `.ia` animations work, so your existing keyed actions translate almost one for one. This part gives the rules, the map, and the Animation Blueprint that plays the clips.

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The anim-instance variables and Thomas's AnimGraph were read from the compiled `ABP_ART_Thomas_Ext` (**verified**). The clip baking and FBX round-trip were tested in Blender (**tested**). Which axis the engine spins `Axles` bones about was not checked (**check**).

1 · Three animation channels

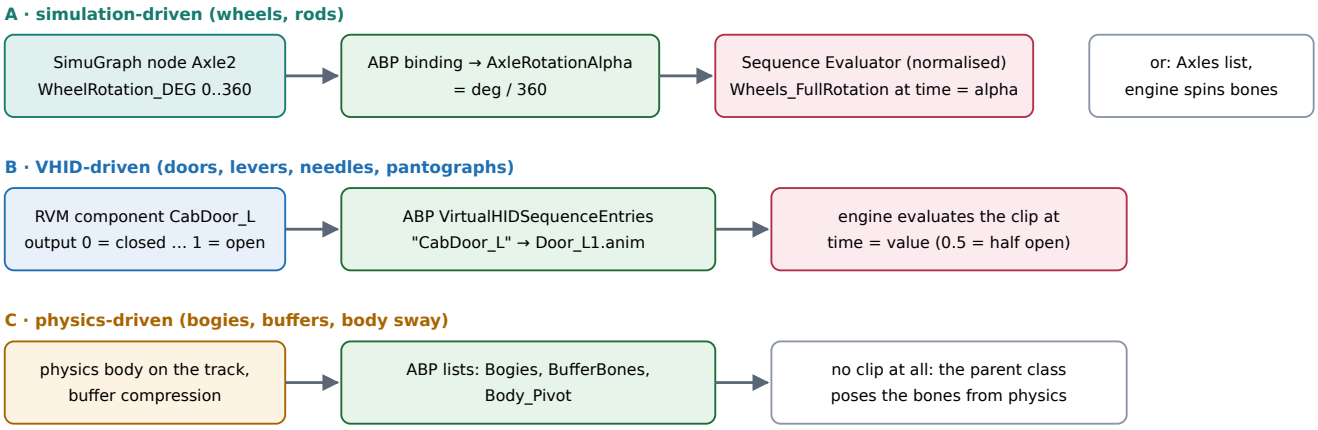


Figure 3.1 · The three channels. You author clips for A and B; C needs only correctly named bones. A clip is never “played”: it is a lookup table from a value to a pose. **verified**

2 · From object keys to a clip: what the add-on bakes

In the master scene a door is an object with rotation keys on frames 0–24. In TSW the door is a bone, and its clip must hold the same motion as pose-bone keys. The add-on samples every frame, computes the object's motion relative to frame 0 in the bone's frame (through any moving parent, so a pantograph's upper arm bakes correctly under its lower arm), and writes location/rotation/scale keys with linear interpolation into an action named after the clip. One action can hold several bones (both pantograph arms share `Pantograph_F`). Each action is exported as its own FBX with the take named after the clip, so the editor imports `Door_L1`, not `Armature|Scene`. **tested**: the baked pose matched the object's world motion to 1e-4 at every sampled frame, including the child-of-bone case, and the clips survived an FBX round trip.

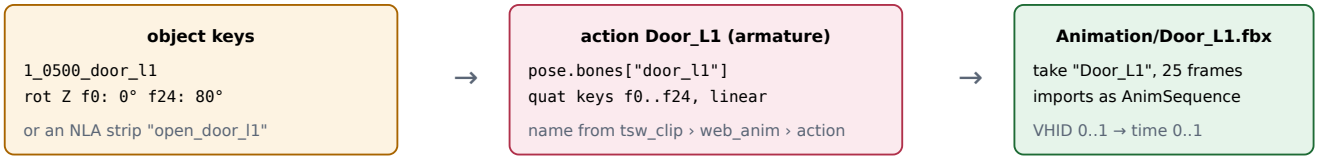


Figure 3.2 · One clip's journey. The clip's length in frames does not matter to the game; only its first and last pose do, because it is scrubbed normalised.

3 · Wheels and rods: Axles list or one-revolution clip

Two mechanisms exist and Thomas uses both: **verified**

Mechanism	How	Use it for
`Axles.Entries` on the ABP	List the axle bones (`axle1, axle2…`). The parent class rotates them from the matching SimuGraph axle every frame; no clip needed.	Diesels, electrics, wagons: plain wheelsets.
`Wheels_FullRotation` clip	A clip holding exactly one revolution of everything that turns with the wheels: wheels, crank pins, rods, pistons. Scrubbed by `WheelRotation_DEG / 360` .	Steam: rods and valve gear; anything geared to the wheels.

The add-on always writes `Wheels_FullRotation` : if the scene has a clip of that name it is baked from your keys (rods included); otherwise it generates one that turns every axle bone 0→360° about its Y axis over 60 frames, forward roll for +X travel (**tested** : the wheel top moves +X). Frame 0 and the last frame are identical poses, as the clip must loop. Which local axis the `Axles` mechanism spins is not documented; the clip route is exact regardless, which is why the guide recommends exporting it anyway. **check**

4 · Doors, levers, needles, pantographs: VHID clips

Every control-driven part is the same recipe: one clip from closed/zero/lowered at frame 0 to open/full/raised at the last frame, and one entry in the ABP's `VirtualHIDSequenceEntries` mapping the RVM component name to the clip. The engine evaluates the clip at time = the VHID's output value.

Part	VHID component (RVM)	Clip	Notes
Cab doors	`CabDoor_L` , `CabDoor_R` (RailVehicleDoorComponent)	`Door_L1` , `Door_R1`	Door state machine outputs 0..1 over `transition_time` .
Throttle, reverser, brake handles	`ThrottleControl` … (IrregularLever)	`Throttle` …	In the **cab** ABP; for a −1..1 lever the clip runs full-back to full-forward and the ABP uses the normalised output.
Gauge needles	a VHID or a sim parameter	`Needle_Speed` …	Thomas keys `CabNeedle_L/M/R` bones; needles driven from a sim value need a small ABP binding like the wheels.
Pantograph	`PantographControl` (PushButton toggle or lever)	`Pantograph_F`	Both arms and the head in one clip; the add-on bakes the whole linkage.
Wipers	`Wipers` (PushButton / WiperComponent)	`wipers`	A cyclic clip is scrubbed by a sawtooth value; the clip should return to rest at the end.
Coupler hook	coupled state	in the coupler's own ABP	Separate small skeletal mesh (ConnectionSkeletalMesh).

> **Tag the component name in Blender**
>
> Give each control-driven object a `tsw_vhid` property with the RVM component name it will be driven by (`CabDoor_L` , `ThrottleControl`). The build sheet then lists the `VirtualHIDSequenceEntries` you have to type: component → clip.

5 · Bogies and buffers: the engine does them

List the bogie bones in `Bogies.Entries` and the buffer bones in `BufferBones` . The parent class yaws bogies to follow the track under the body and translates buffer bones by the compression the physics component computes from the buffer stiffness. No clip, no keys. TSC's rule that the XML pivot wins is replaced by the bone position: put the bogie object's origin at the real pivot before exporting. **verified** (Thomas's lists; bogies empty on a rigid-frame engine).

6 · Wheels and valve gear in three games

	Train Sim Classic	Transport Fever 3	Train Sim World
Wheels	Game rotates `boNNwhNN` nodes from speed and radius; no animation authored	`wheels` event animation: one turn keyed, game loops it by distance	Engine spins `Axles` bones from the SimuGraph axle, or scrubs the one-turn clip
Rods / valve gear	Named rod nodes driven by the game's valve-gear system from wheel angle	Keyed in the same `wheels` animation	Keyed in `Wheels_FullRotation` , scrubbed by the axle angle
Authoring	Nothing	One cyclic action	One cyclic action (or none for plain wheelsets)
Shared source	One action in the master named `Wheels_FullRotation` (or tagged `tsw_clip`) serves TF3's wheels strip and TSW's clip; TSC ignores it.		

7 · The TSC → TSW animation map

TSC animation (.ia, driven by)	TSW	Translation
Door animation ← control value	Door clip ← RailVehicleDoorComponent	Same keys; frame 0 closed.
Pantograph ← `PantographControl`	Pantograph clip ← VHID	Same keys, whole linkage in one clip.
Cab lever ← Regulator / Reverser / TrainBrakeControl	Lever clip ← IrregularLever (cab ABP)	Same keys; value map on the RVM decides notches.
Needle ← speedometer control value	Needle clip ← sim parameter binding or VHID	Same keys; needs a binding in the cab ABP.

TSC animation (.ia, driven by)	TSW	Translation
Wipers ← Wipers control value	Wipers clip ← Wipers VHID	Same keys.
Trolley pole ← custom control values (Lua)	Pole clip ← a custom VHID; logic in SimuGraph / event graph	Keys translate; the Lua becomes Blueprint (Part 5 §7).
Wheel nodes (no animation)	Axles list / Wheels_FullRotation	Generated by the add-on.
Bogie pivots from XML	Bogies list	Bone position is the pivot.
Lights (control values → material)	Headlights VHID → LinearElectricLight → light/emissive	No animation; Part 6.

8 · The Animation Blueprint (RailVehicleAnimInstance)

Every skeletal mesh on the RVV uses an Animation Blueprint whose parent class is `RailVehicleAnimInstance`, not the plain AnimInstance. The parent adds four lists and a graph node that does the work. Thomas's entire exterior ABP: **verified**

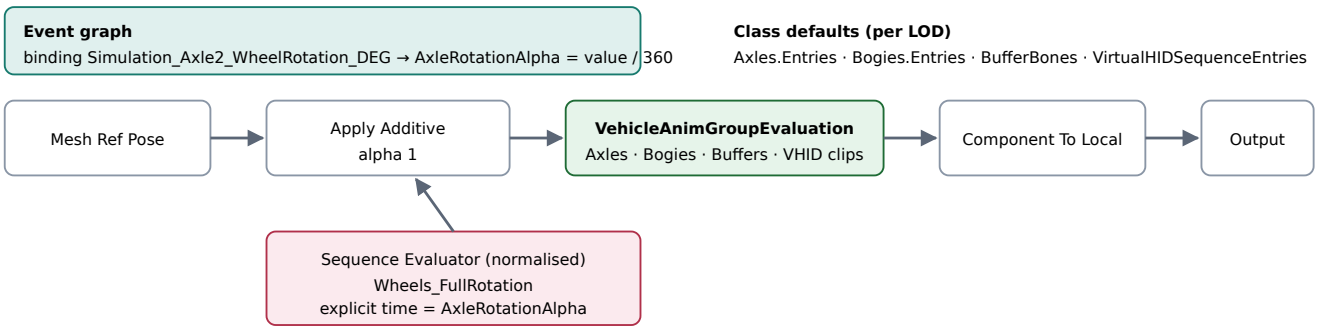


Figure 3.3 · The whole exterior ABP. Five graph nodes, one variable, one binding. The cab ABP is the same with lever clips in its VHID entries and no wheel evaluator.

Parent-class property	Meaning / Thomas value
`Axles.Entries[]`	Bones the engine spins from axle rotation: `axle1, axle2, axle3, axle_Rod_L, axle_Rod_R`.
`Bogies.Entries[]`	Bones yawed to follow the track. Empty on Thomas.
`BufferBones[]`	`buffer_fl, buffer_fr, buffer_bl, buffer_br`.
`VirtualHIDSequenceEntries[]`	Per LOD: {Component (RVM VHID name), Sequence}. `CabDoor_L` → `CabDoor_L.anim`, `CabDoor_R` → `CabDoor_R.anim`.
`CachedModelActor`	Runtime pointer to the RVM, resolved for you.

The simulation value arrives through a **property binding**: a variable of the struct type {NodeReference("Simulation", "Axle2"), ParameterReference("WheelRotation_DEG")} that the parent class evaluates every frame. The build sheet lists the axle names so you know which node to bind.

> **What the wagon stream adds about the lists**
>
> On creating the ABP (parent `RailVehicleAnimInstance`) the editor **auto-filled** four Axles entries and two Bogies entries from the bone names. The wheels still did not turn in Play until two more things were set: the ABP's **model actor** reference (the wagon's RVM class), and, on *each* Axles and Bogies entry, the simulation component (`Simulation`) and the SimuGraph node it belongs to (bone `Axle1` ↔ node `Axle1`, bone `Bogie1` ↔ node `Bogie1`). So an entry is "bone + sim node", not just a bone. After that, wheels and bogies moved with no clip at all. **community**

9 · Keying rules and troubleshooting

Rule	Because
Frame 0 = rest (closed, lowered, zero); last frame = fully open/raised/full	The clip is scrubbed 0..1 by the value.
Linear interpolation, no overshoot	A value of 0.5 must look half-way; eased keys make the HUD and the mesh disagree.
Wheels_FullRotation starts and ends on the same pose	It wraps at 360°.
One clip per control, named like the control	The VHID entry maps one component to one sequence.
Keep object-level keys in the master; never key the preview rig	The rig is rebuilt from the master on every export.

Symptom in the editor	Cause and fix
Door swings about the wrong point	Object origin not on the hinge in the master.
Part jumps when the vehicle spawns	Frame 0 is not the rest pose; or the clip was keyed from an already-open state.
Wheels turn backwards	Clip direction; re-export with the generated clip, or flip the sign of the keyed rotation.
Clip imports named "Armature|Scene"	Exported by hand without renaming the scene; the add-on names the take after the clip. Rename the asset in the editor.
Pantograph head stays behind when the arms rise	The head was weighted to Body_Pivot: it must be a child of the upper arm in the master.
Lever moves but the HUD value does not	The ABP entry points at the view hitbox name, not the RVM component name.